A thesis submitted in partial fulfilment of the requirements for the degree of Master of Science (MSc) in:

Data Science

UNIVERSITY COLLEGE LONDON

Modelling high performance athletes for 100 m

Author:
Candidate Number: MSJP5

Supervisor: Prof J. Griffin

Department of Statistical Science
UNIVERSITY COLLEGE LONDON

Date of submission: September 7, 2025
Word count: 9836

Contents

Abstract **3**

1 Introduction **3**

1.1 Background . . . 3

1.2 Aim of thesis . . . 3

2 Literature Review **5**

2.1 Overview . . . 5

2.2 Statistical Models for Athletic Performance . . . 5

2.3 Modern approaches . . . 5

2.4 Random Effects Models . . . 6

2.5 Fixed Effects Models . . . 6

2.6 Combined Effects Models . . . 6

2.7 Random Walk Models . . . 6

2.8 Key Innovations . . . 7

2.9 Gaps in Current Research . . . 7

3 Methodology **8**

3.1 Data Description . . . 8

3.2 Model Methodology . . . 9

3.2.1 Handling Missing Data . . . 9

3.2.2 Predictive Questions Methodology . . . 9

3.2.2.1 What age to decide funding? . . . 10

3.2.2.2 How well is an athlete compared to peers of the same age? . . . 10

3.2.2.3 Who's likely to reach a championship final? . . . 10

3.2.3 Random Effect Model . . . 11

3.2.4 Fixed Effect Model . . . 12

3.2.4.1 Predictive Distribution . . . 14

3.2.5 Combined Effects Model . . . 16

3.2.5.1 Predictive Distribution . . . 17

3.2.6 1st Order Random Walk Model 18
3.2.7 2nd Order Random Walk Model 19

4 Results **20**
4.1 Model Evaluation Metrics 20
4.2 Out-of-Sample Prediction Results 21
4.3 Results Analysis 22
4.4 Predictive Questions Results 24
4.4.1 When age to decide funding? 24
4.4.2 How well is an athlete compared to peers of the same age? 26
4.4.3 Who's likely to reach a championship final? 27

5 Discussion **29**
5.1 Strengths of the study 29
5.2 Limitations and extensions 29

References **31**

Appendix **34**
A.1 Random Effect STAN model 34
A.2 Fixed Effect STAN model 35
A.3 Combined Effect STAN model 36
A.4 1st Order Random Walk STAN model 37
A.5 2nd Order Random Walk STAN model 38
A.6 Posterior covariance matrix for Male 100 m 39
A.7 Additional results for Female 100 m 39

Abstract

Forecasting high performance in athletics is an important challenge in sport science, with implications for talent identification, funding decisions and competition strategy. Previous studies often rely on all recorded performances, but this can hide an athlete's true long term progress as performances can be mixed up with short term fluctuations due to injuries or race conditions. This dissertation instead models athletes' season's best 100 m performances, providing a clearer signal of long term progression. Competition data from 672 male and 518 female sprinters aged 17-30 were analysed using Bayesian hierarchical models. Fixed effect quartic polynomials, multivariate normal random effects, and random-walk priors were implemented, with Bayesian inference using MCMC to estimate parameters and produce predictive distributions. Model evaluation shows that fixed effects models outperform more complex models, and so offer better calibration and accuracy. Predictive problems to answer include identifying when athletes should receive funding, benchmarking performance against peers and forecasting finalists for major championships. The study establishes season's best based Bayesian modelling as a practical and flexible framework, delivering insights for coaches, sponsors, and sporting organisations.

1 Introduction

1.1 Background

The modelling of high athletic performance has recently been an interesting topic for sports scientists, analysts and sponsors due to its impact on talent spotting, funding decisions, resource allocation and other applications. Accurate forecasting of athlete potential can allow coaches, governing bodies and sporting organisations to prioritise training programs and allocate funding accordingly to athletes who are most likely to reach world stage success.

Historically, studies of athletic progression have relied on longitudinal models that include every competitive result without distinguishing between variability due to training cycles and genuine changes in underlying athletic capability. However, these methods may mix up long-term trends with short-term noise, which can make an athlete's underlying performance trajectory less clear.

1.2 Aim of thesis

This study addresses these challenges by focusing on the athletes' season's best performance as the measure of interest. Since traditional models utilise every recorded performance, a season's best approach captures the athlete's peak performance within each competitive year. This offers a more robust and interpretable representation of an athlete's underlying performance level, which will be our target variable. We can capture more accurate trajectories for elite performance forecasting by separating out peak annual results, which reduces the impact of the unpredictability in athletics such as injuries, motivational swings, and suboptimal competition settings [1].

Our goal is to predict season's best performance using Bayesian hierarchical models. An illustrative prediction based on the data is presented in Figure 1. The graph shows the observed season's best performances and a fitted career performance trajectory with 95% credible interval for a male 100 metre sprinting athlete. In this example there is evidence of a trend where

his performance level improves from 10.8 seconds, at age 19, to 10.3 seconds, at age 21, remains steady around the 10.3 second mark up to age 26 and then shows a decline in performance after this age. We aim to model such behaviour for all athletes, separately for males and females, in order to predict the season's best performance at future ages. In doing so we seek to address broader predictive questions like: determining the optimal timing for funding for athletes that are projected to reach elite performance, comparing an athlete's relative performance to peers of the same age and predicting which athletes will become elite performers in future major competitions, such as the Olympics or World Athletics Championships.

The Bayesian hierarchical models implemented in this project include fixed effects, random effects and random walks. Bayesian inference methods, in particular Markov Chain Monte Carlo (MCMC), are employed to quantify uncertainty in predictions and generate probabilistic estimates for athletes' potential in achieving elite performance status in the future. We also use this to compare the performance of these different models to see which approach is most effective for prediction.

Overall, this thesis contributes a novel approach to sports performance modelling, by employing season's best performances within a Bayesian hierarchical model framework. This enables clearer identification of athletic potential, appropriate handling of performance variability and better informed insights for decision makers such as coaches within elite sporting organisations.

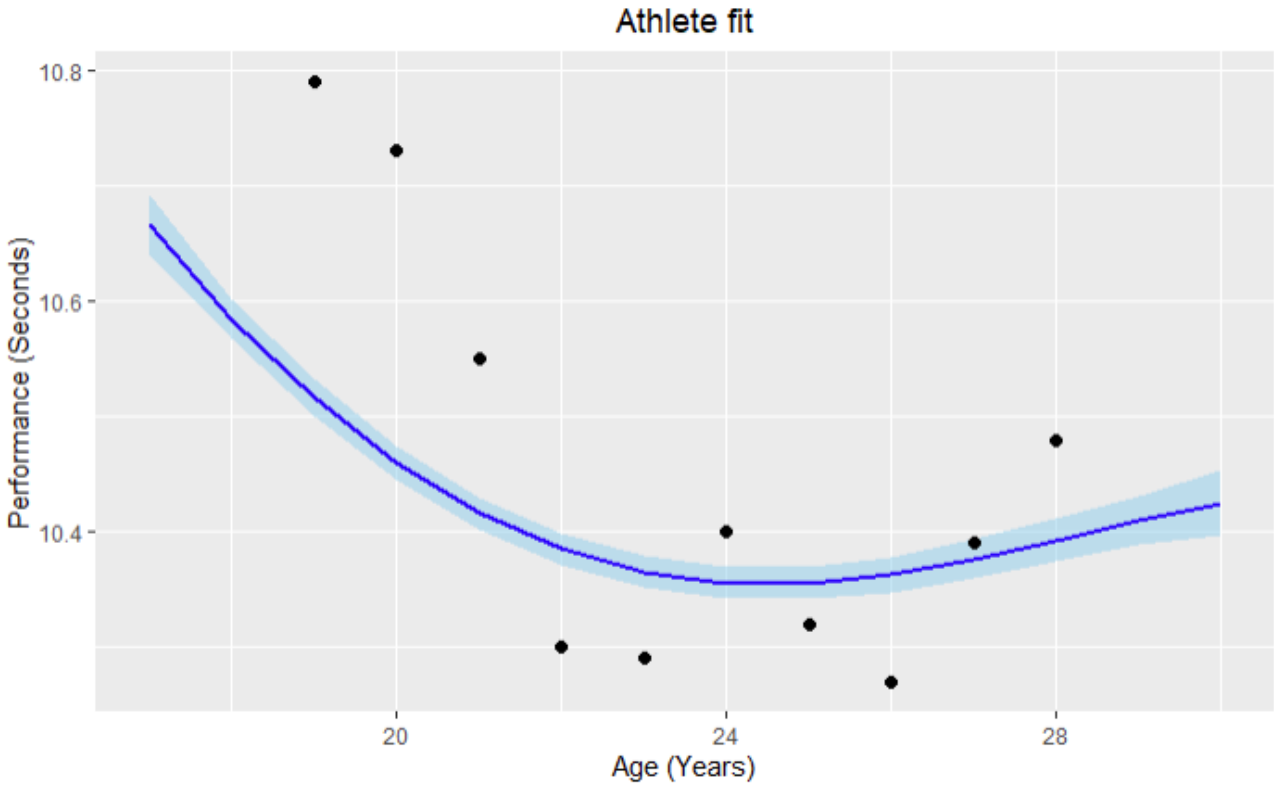

Figure 1: Observed performances (black dots) of a male 100 metre athlete. The trajectory is shown as the posterior median (dark blue line) of a fitted age polynomial and 95% credible interval (light blue shade)

2 Literature Review

2.1 Overview

Predicting athletic performance, particularly in events like the 100 m sprint, can be a complex challenge. These predictions can have significant implications for talent identification, funding allocation and training optimisation. This literature review explores the various methodologies for athletic and sporting performance, focusing on the different statistical modelling such as random effects, fixed effects and random walk models. The following will explore an in-depth analysis of major contributions to the field and where current research lies.

2.2 Statistical Models for Athletic Performance

Early research on athletic performance modelling typically relied on cross-sectional or basic longitudinal analyses, such as least-squares regression or simple average comparisons by age group. For example, Born et al. [2] compared different approaches: following different athletes at a single point in time and following the same athletes over multiple years. They found that cross-sectional data produced slower estimated progression, especially in lower percentile performers, and significantly underestimated trajectories toward peak elite performance in high-achieving athletes. They rarely distinguish between regular competition performances and peak performances. This can obscure true talent progression, as short-term variations, due to training cycles, injuries or competition conditions, are averaged into the signal of long-term development [3].

2.3 Modern approaches

Modern research employs longitudinal and hierarchical statistical models. These allow for individual-level inference, can accommodate missing data naturally, and can explicitly model both fixed effects (shared across athletes) and random effects (athlete-specific deviations). They naturally quantify uncertainty and integrate prior knowledge with observed data [4]. Bayesian methods, particularly MCMC, have become instrumental in sports analytics due to their ability to provide probabilistic interpretations of athlete performance, which addresses the uncertainty inherent in longitudinal data [4].

Berry, Reese, and Larkey [5] were early proponents of Bayesian hierarchical models in sports, splitting performance into individual, season, and age effects, with individual ageing functions. These are mathematical representations of how an athlete's performance changes with age and are important because athlete careers tend to follow a non-linear trajectory. Performances generally improve during late teens, peak in early to mid twenties and gradually decline with age.

Research by Berthelot et al. [6] applied Bayesian hierarchical modelling to analyse athletic performance across age, providing insights into the peak performance age and the subsequent rate of performance decline. This research emphasised the effectiveness of Bayesian hierarchical models in accurately capturing age-performance dynamics, particularly for elite athletes. They are well suited since they estimate population-level ageing patterns as well as allowing each athlete to deviate from the average curve. These models borrow strength across athletes to stabilise and inform estimates and can produce individualised trajectories. Moreover, the

Bayesian framework quantifies uncertainty in forecasts, which is useful for evaluating models.

Later work by Brander, Egan, and Yeung [7], Bradbury [8], and others introduced polynomial or spline-based models for both population and individual ageing functions. Polynomial functions are a simple way to describe the general rise and fall in performance with age. Splines join together smooth curve segments, which makes them more flexible in following the exact shape of an athlete's career path. These can make models better reflect reality as they can tackle the irregularity of how performances change over time.

2.4 Random Effects Models

Random effects models have been widely used in sports analytics due to their ability to model variability among individuals. A random effect model assumes that the observed performances of athletes are random deviations from an overall population mean trajectory, which is essential for modelling the individuality of athlete development [5]. It is also useful for longitudinal data where Bates et al. [9] demonstrated the effectiveness of random effects models to capture athlete-specific variability and improve predictive accuracy for longitudinal datasets, highlighting their suitability for athletic performance data.

2.5 Fixed Effects Models

Fixed effects models capture average, population wide trajectories, assuming the same relationship for all athletes. They describe the average performance trajectories across age. This is commonly done via polynomial regression [3] and have shown high accuracy in capturing non-linear age-performance relationships observed in athletes, making it standard practice in sports performance analysis [10].

2.6 Combined Effects Models

Combined or mixed effects utilise both fixed and random effects which allows for more precise inference even for athletes with sparse data. Mixed effect models are widely used in sport analytics as they allow for refined estimation of performance curves and improved forecasting [11][12]. They are popular for their flexibility and robust predictive capabilities in sports performance research. Hopkins et al. [13] highlighted that mixed effects models provide superior accuracy in predicting athletic outcomes due to their capacity to handle complex hierarchical data structures, which are commonly found in athletic data.

2.7 Random Walk Models

Random walk models treat each athlete's performance as a stochastic process where future outcomes are best predicted by their most recent performances. In their simplest form, random walk priors allow each successive year's season's best to be modelled as the previous year's performance plus a normally distributed term [14].

A notable application is provided by Terpstra and Schauer [15], who applied a one-sided random walk model, with a mixture distribution for the error term, to predict track and field world records, of which included the men's 100 m sprint. They grounded the inference in likelihood

based methods and used MCMC to produce prediction intervals. Although focused on world records rather than season's bests, this method highlights how random walk structures can effectively model extreme performance outcomes over time.

2.8 Key Innovations

An important step forward in methodology has been the use of an athlete's season's best performance as the main indicator of their underlying ability [3][1]. This approach helps to limit the impact of short term factors such as injury or unfavourable conditions, and provides a more reliable measure of peak performance in a given year.

However, this choice involves a trade-off. By focusing only on the best result in a calendar year, we lose information contained in other performances throughout that year, so we lose observations. There is strong evidence of within-season effects on athletic performance [11] so although ignoring this information simplifies modelling there is a risk in overlooking meaningful patterns in athlete performance.

More recent work has introduced methods that expand the modelling framework. For example, models incorporate covariates, such as wind speed or the competition level. Other approaches introduce non-linear trajectories across age or heavier-tailed error distributions to better reflect observed performances [11][12].

This highlights a key motivation for the thesis, that although season's bests could be used as a performance indicator, there is a need to evaluate how well season's best based models can answer applied predictive questions. These include when to fund athletes, how to benchmark athletes against peers and which athletes are most likely to reach major competitions, despite the simplification in the framework.

2.9 Gaps in Current Research

While considerable progress has been made, several gaps remain to be addressed. One of these challenges in the field is accurately managing and modelling missing data. Many studies opt for simple exclusion or rudimentary imputation techniques, potentially biasing results (Little & Rubin [16]). Improved approaches such as direct modelling of observed data with indexing methods remain underutilised.

Another limitation is the sparse exploration of individual athlete predictive modelling beyond aggregated population-level predictions. While hierarchical models provide individual level predictions, further work could refine these predictions by incorporating more detailed athlete-specific data, such as training load, injury history, and psychological factors (Borresen & Lambert [17]).

There is also a lack of research dedicated to optimising decision making timelines for athlete investment based on predictive outcomes, which is critical for sporting organisations' strategic planning and resource allocation.

3 Methodology

3.1 Data Description

Our analysis uses 100 m sprinting competition results from the World Athletics website. The data consists of performances from both male and female sprinters in 100 m races, recorded between February 2001 and August 2021. Performances are taken from official competitions and include details such as finishing time, wind conditions, venue, date, athlete nationality, date of birth and competitor age at the time of performance.

In total, the dataset covers 2,978 unique athletes, providing a large and diverse sample of sprinters over a twenty-year period. In total there are 114,534 observed performances, split across 65,535 male performances and 48,999 female performances. Athletes span an age range from around 12 to 45 years old, although the majority of performances are concentrated between the late teenage years and the early thirties. Performance levels differ between genders. For men, 100 m times in the dataset range from 9.74 seconds to 11.00 seconds, while for women, times range from 10.61 to 12.30 seconds.

The dataset is longitudinal, where each athlete has multiple performance results across different seasons. This repeated structure allows the tracking of individual development over time.

Given the purpose of the study, to model athletes' season's best performances, the dataset was processed to include only the single best performance result recorded for each athlete in each competitive season. Typically, an athletic season aligns with a calendar year, from January 1st to December 31st [11], making it straightforward to standardise performance records across athletes. To ensure consistency and to simplify the continuous nature of age, each athlete's age was defined as their integer age on January 1st of the given competitive season. Using an integer-based age definition, however, involves a trade-off. Athletes born at the very end of the calendar year, e.g, December 31st will differ in assigned age by a full year to someone born near the start of the calendar year, e.g January 1st, when there is only a minimal chronological age difference. Although this approximation introduces slight biases around year-end ages, it is expected these effects are unlikely to significantly affect the analysis. This approach is common practice within performance analytics and is generally accepted in the literature due to its practical ease and interpretability, [3]. The minor approximation introduced is justified by the simplified and substantial analytical advantages gained.

The full dataset initially consisted of performances from 1944 and 1297 male and female athletes respectively. However, to ensure numerical stability in the models and to focus on the specific research questions about high-performing athletes, further filtering was applied. First, a season's best for an athlete in a given year was only retained if the athlete had at least three recorded performances in that season. This requirement helps to ensure that the selected season's best is a reliable indicator of the athlete's peak capability for that year, as relying on just one or two results could lead to the inclusion of outliers or performances influenced by unusual conditions e.g., extreme weather or injury. Additionally, only athletes with at least four years of season's best data were included. This condition increases the predictive power of the models, particularly the random effect models, which benefit from multiple observations to estimate each athlete's unique performance trajectory more accurately. Since the focus is on high performance, the dataset was further restricted to male athletes whose all-time personal best (across all years) was faster than 10.5 seconds, and for female athletes, faster than 11.6 seconds [11]. This cut-off ensures that the analysis is focused on athletes with the potential to compete at the highest levels. Finally, to manage numerical stability and computational

feasibility, especially when estimating the large covariance matrices involved in the random and fixed effect models, the analysis was limited to performances recorded between the ages of 17 and 30. Including a wider age range would significantly increase the dimensionality of the covariance matrices, which raises both computational demands and the risk of numerical instability, without much additional insight.

After these changes, the final dataset for male athletes had 672 athletes and for female there were 518 athletes.

3.2 Model Methodology

This study uses Bayesian hierarchical models to describe and predict athletes' season's best performances across different ages. The hierarchical structure allows for individual athlete variability while capturing population level effects. All model parameters, including polynomial coefficients and the covariance structure, are estimated using Bayesian inference. This section outlines these modelling approaches with a particular emphasis on the predictive distributions. This is interesting to us since the goal is to not only capture past trajectories but to make informed forecasts about future performance. Predictive distributions provide a full probabilistic description of possible performances in future ages, not just a single age prediction. This also means uncertainty can be quantified and the applied questions, we seek to answer, can be addressed.

3.2.1 Handling Missing Data

Due to the longitudinal data we are using, where we have multiple performance results for each athlete, not every athlete will have a season's best available for every age within the considered age range. Therefore we naturally face the problem of missing data. Explaining this further, suppose that $y_{i,a}$ represents the season's best for athlete i at age a, where the youngest and oldest ages in the dataset are represented as A and B, respectively. Our dataset contains missing values for certain athlete-age combinations because most athletes do not compete or achieve a season's best (since we require a minimum of 3 performances per year to obtain a season's best) at every single age from A through to B.

To address this, we only model observed performances as opposed to imputing the missing performances since imputing unobserved season's bests would require strong and potentially unrealistic assumptions about athlete trajectories, risking biased inferences. To do this we represent each athlete's observed performances using an indexing approach, whereby each athlete i has a corresponding vector age_idx[i] indicating the specific ages at which those performances were observed for athlete i. This indexing allows direct modelling for only the available data points and thus effectively circumvents the missing data.

3.2.2 Predictive Questions Methodology

A key motivation for developing the models in this dissertation is to answer practical predictive questions that are relevant to coaches, sponsors, and sport governing bodies. Here we will discuss how the modelling framework and MCMC can be applied to such questions.

3.2.2.1 What age to decide funding?

The first question we want to answer is: for a given athlete at what age should funding be decided? A critical problem in high performance sport is determining when to classify an athlete as high potential and, therefore, eligible for targeted investment. If we consider an athlete who currently records a 100 m time of 10.20 seconds at age a, we would like to know at what age we are confident that this athlete reaches 'elite' status.

Using MCMC samples from a posterior distribution and for a particular athlete i we would like to find the age that yields the highest probability of reaching elite status. For a threshold time T we want to calculate the probability that the posterior season's best time, at an age a, is less than T and then repeat this for all ages greater than the last observed age for that athlete, A^*. This can be shown by:

$$P(y_{i,a} < T \,|\, y_{i,1}, \ldots, y_{i,A^*}) \,, \qquad a > A^* \tag{3.1}$$

This probability provides a measure of whether the athlete is on track to reach elite performance, for that particular age.

3.2.2.2 How well is an athlete compared to peers of the same age?

For the second predictive question, sponsors or coaches are often interested in how well an athlete compares relative to peers of the same age. If we consider a particular athlete i at a fixed future age, a, and we want to compare performance with another athlete, k, then we compute:

$$P(y_{i,a} < y_{k,a} \mid \boldsymbol{y}) \tag{3.2}$$

This is the posterior conditional probability that athlete i outperforms athlete k at age a. But this can also be extend by finding the probability that a specific athlete i is faster than all others at a fixed age a, so comparing an athlete relative to all other athletes in the same age cohort. Now the probability of interest becomes:

$$P(y_{i,a} < y_{k,a} \;\; \forall\, k \neq i \mid \boldsymbol{y}) \tag{3.3}$$

which is the posterior conditional probability that athlete i outperforms all others at age a. This can provide a probabilistic ranking of athletes for a particular future age. For example, among all athletes aged 23 in the last observed season, we can identify those most likely to outperform their peers both now and in subsequent years. These rankings can be used to identify young future talent. Furthermore, instead of relying on raw times, selectors can use probabilities to prioritise funding, training camp invitations or specialist coaching.

3.2.2.3 Who's likely to reach a championship final?

The final predictive task is to forecast which athletes are likely to qualify as finalists in upcoming major championships or Olympics. This is because final appearances are one of the most visible and consequential markers of elite status in athletics. Reaching a major final is an important indicator for global competitiveness. So the probability we wish to find is:

$$P(\text{athlete } i \text{ in top 8 at age } a_i^* \mid \boldsymbol{y}) \tag{3.4}$$

where a_i^* is the age athlete i will be at the championship year.

3.2.3 Random Effect Model

To capture individual differences in athletic performance trajectories, we first consider a random effect model that ignores fixed effects and models each athlete's season's best performances as deviations from an overall mean trajectory, governed by correlated random effects across ages. This served as our initial baseline model.

Let $s_{i,a}$ be an indicator, where

$$s_{i,a} = \begin{cases} 0, & \text{if } y_{i,a} \text{ is observed} \\ 1, & \text{if } y_{i,a} \text{ is missing} \end{cases}$$

Only observed $y_{i,a}$ values are included in the likelihood

$$y_{i,a} \mid \gamma_{i,a}, \sigma_y \sim N(\gamma_{i,a}, \sigma_y^2), \qquad \begin{array}{l} i = 1, 2, \ldots, N, \\ \text{if } s_{i,a} = 0 \end{array} \qquad a = A, \ldots, B$$

where $\sigma_y \sim$ Half-Cauchy(0, 2.5) represents the residual standard deviation.

Let J be the total number of ages used. Since we consider only ages from $A = 17$ to $B = 30$ then $J = 14$. Each athlete has a J-dimensional random effect vector: $\boldsymbol{\gamma_i} = (\gamma_{i,A}, \ldots, \gamma_{i,B})$ representing deviations from the overall mean trajectory at each age. It is assigned a multivariate normal prior:

$$\boldsymbol{\gamma}_i \sim N(\boldsymbol{\gamma}_0, \boldsymbol{\Sigma}), \tag{3.5}$$

where $\boldsymbol{\gamma_0} \in \mathbb{R}^J$ represents the mean random effect across all athletes i.e the overall mean random effect. $\boldsymbol{\Sigma} \in \mathbb{R}^{J \times J}$ is the covariance matrix associated with the gammas, and captures correlations among these random effects at different ages, using a correlation matrix, $\boldsymbol{\Omega}$.

The correlation matrix, $\boldsymbol{\Omega}$, takes an LKJ correlation prior, recommended by [18]. It is widely recommended to use in hierarchical models because it gives a simple and flexible way to model correlations. With the hyper-parameter $\eta = 1$, the LKJ prior, $\boldsymbol{\Omega} \sim \text{LKJ}(\eta = 1)$, is uniform. This means all correlation structures are equally likely. Higher values of η push the prior to the identity matrix and lower values (< 1) favour stronger correlations [4]. Alternative priors, such as the inverse-Wishart distributions, are less flexible, since this prior is put directly on the whole covariance matrix, as opposed to separating the priors on the scales and correlation matrix. The separation of variances and correlations makes the model more realistic and improves stability in estimation [4] [19]. A half-Cauchy prior is used on the scales, σ_j which are also the marginal standard deviations. This was chosen since it is a weakly-informative prior and effectively used together with the LKJ prior [18].

Priors for the model parameters can be mathematically shown as:

$$
\begin{aligned}
&\boldsymbol{\gamma_0} \sim N\big(0, 10^3 \boldsymbol{I}_J\big), \\
&\boldsymbol{\Sigma} = D_\sigma \boldsymbol{\Omega} D_\sigma, \\
&\sigma_j \sim \text{Half-Cauchy}(0, 2.5), \qquad j = 1, \ldots, J, \\
&\boldsymbol{\Omega} \sim \text{LKJ}(1)
\end{aligned}
\tag{3.6}
$$

D_σ is the $J \times J$ matrix with σ_j on the diagonal and zeros elsewhere, and $\boldsymbol{\Omega} \in \mathbb{R}^{J \times J}$ is the correlation matrix. $\boldsymbol{\sigma}$ is the scale vector of the prior scales σ_j's: $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_J)$ and are used to get the diagonal matrix needed for D_σ in equation 3.6.

This random effect model allows each athlete to have their own trajectory of season's best performances across ages, but does not include population-level fixed effects. All deviations are modelled as coming from a shared multivariate normal structure, allowing for borrowing of strength across athletes and ages. Only observed values are included in the likelihood, with missing entries ignored (as indicated by $s_{i,a}$).

This random effect model serves as a baseline for comparison with more complex models (such as those including fixed effects or random walks), and highlights the importance of accounting for individual variation and age-age correlations in modelling athlete performance data.

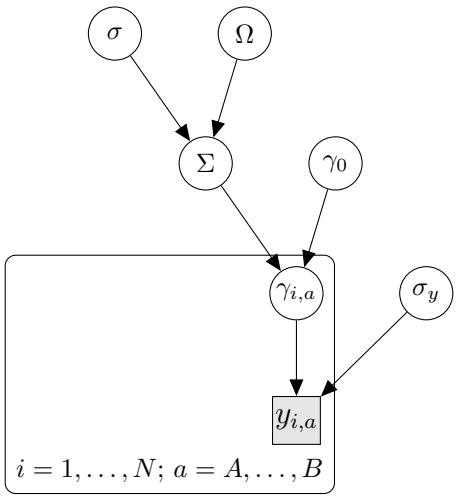

Figure 2: A directed acyclic graph (DAG) of hierarchical model structure for Random Effect model

3.2.4 Fixed Effect Model

We consider a fixed effect model to capture the overall progression of athletic performance across ages. In this model, each athlete's observed season's best performance is modelled from a common population level trajectory, without explicitly modelling athlete-specific deviations from this mean curve.

Mathematically, for athlete i at age a, the model is:

$$y_{i,a} = f(a) + \epsilon_{i,a}$$

where:

- $f(a)$ is the fixed effect, representing the population-average performance at age a, centred at the average from the age range used

- $\epsilon_{i,a}$ are correlated errors, allowing for residual dependence across ages within an athlete's career

$$f(a) = \beta_0 + \beta_1(a - C) + \beta_2(a - C)^2 + \beta_3(a - C)^3 + \beta_4(a - C)^4 \tag{3.7}$$

$$\beta_k \sim N(0, 100), \quad k = 0, 1, 2, 3, 4. \tag{3.8}$$

Equation 3.7 is a 4th order polynomial centred at an average age, C, with an intercept term. Here ages, a, range from 17 to 30 and C=23.5, which is the average of this age range. β_0 is the intercept term and is interpreted as the average population performance at the centred age, C. β_1 represents the rate of change of performance per age year. The higher order terms add flexibility and capture the non-linear patterns.

A quartic polynomial was chosen since the relationship between athletic performance and age is known to be non-linear. Performance tends to improve rapidly during late teens and early twenties, reach a plateau or peak in their mid to late twenties, and then decline with increasing age [20]. To capture this trajectory, we model the population-level mean performance as a polynomial function of age. To determine the appropriate degree for the polynomial, exploratory data analysis was first conducted. The average season's best performance for each integer age was calculated and then plotted, see Figure 3.

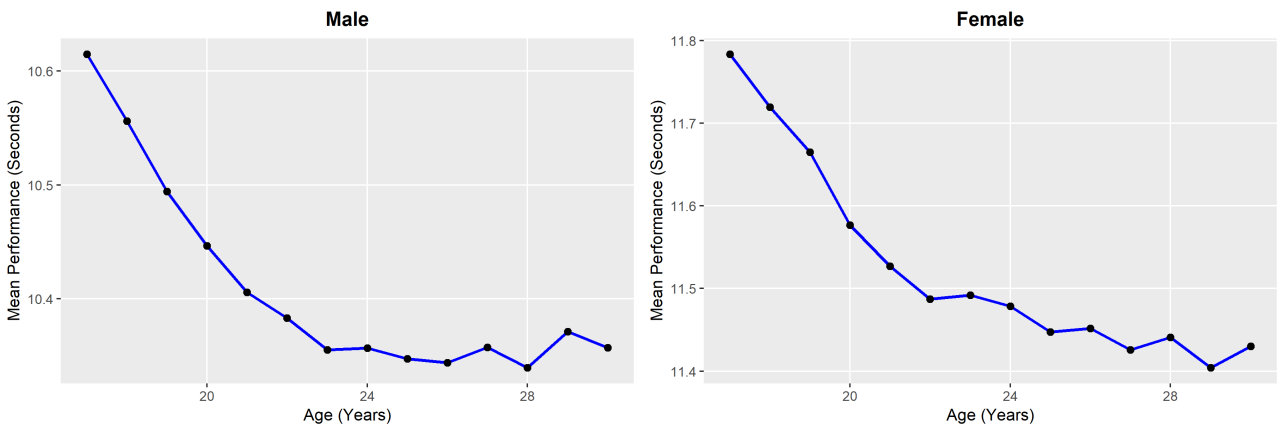

Figure 3: Average male and female performance by age

Polynomial regression models were then fitted, using least squares with varying degrees of polynomial which were centred around an average age and included an intercept term. For each model, the model fit was examined using information criteria AIC and BIC. These balance model fit against model complexity, penalizing overfitting [21]. The results showed that, the quartic polynomial (degree 4) provided a noticeably improved fit, as indicated by lower AIC and BIC values. Therefore, for both empirical and practical reasons, a quartic polynomial will be used as the fixed effect. Furthermore, the coefficient estimates obtained from this exploratory regression, were used to initialise the polynomial parameters in the Bayesian fixed effect model, which helped ensure stability and efficient model fitting. This degree of polynomial is well established in the literature for modelling age-related performance curves [7].

This model implements the fact that repeated performances, from the same athlete, are not independent, which means the errors are correlated across ages for each individual and the

correlation structure is shared across all athletes. The likelihood for each athlete is constructed as a multivariate normal over their observed performances:

$$\boldsymbol{y_i} \sim N(\boldsymbol{f(a)}, \boldsymbol{\Sigma}_{M_i,M_i}), \qquad i = 1, 2, \ldots, N \qquad a = A, \ldots, B$$

where the mean is the population polynomial vector of ages, $\boldsymbol{f(a)}$, and a covariance matrix $\boldsymbol{\Sigma}$ which is shared by all athletes. M_i are the observed indices of ages for athlete i since in the modelling only observed performances are used. Therefore, the covariance used is the corresponding sub-matrix of $\boldsymbol{\Sigma}$ for the ages observed for that particular athlete. Since the likelihood is a multivariate normal, the observed performances for each athlete i form a vector $\boldsymbol{y_i}$ with mean $\boldsymbol{f(a)}$ and covariance matrix $\boldsymbol{\Sigma}_{M_i,M_i}$. The model is structured like this because we assume the performances, that are observed at different ages, are not independent but correlated. We treat them jointly as a vector to capture the covariance across ages within an athlete. The priors for this covariance matrix are the same as the priors for the random effect, $\gamma_{i,a}$, described in Section 3.2.3:

$$\begin{aligned} &\boldsymbol{\Sigma} = D_\sigma \boldsymbol{\Omega} D_\sigma, \\ &\sigma_j \sim \text{Half-Cauchy}(0, 2.5), \quad j = 1, \ldots, J; \\ &\boldsymbol{\Omega} \sim \text{LKJ}(1) \end{aligned} \tag{3.9}$$

The hierarchical structure can be shown in Figure 4.

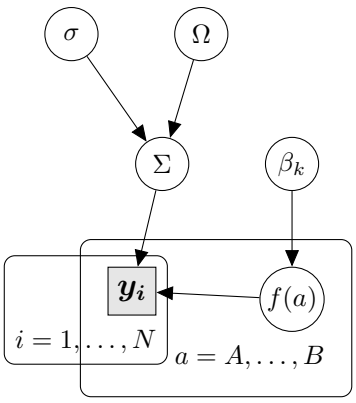

Figure 4: DAG of Fixed Effect hierarchical model

3.2.4.1 Predictive Distribution

An advantage of this fixed effect model is that predictions for an athlete's future, unobserved season's best can be found conditional on their observed results. This is because under the model, an athlete's vector of season's best performances, across all ages, is assumed to follow a multivariate normal distribution, with the mean determined by the fitted population trajectory and the covariance capturing correlations across ages.

For a given athlete i, we observe season's best performances up to age a_i. Let $y_{i,1:a_i}$ be these observed values and let $y_{i,(a_i+1):B}$ represent the unobserved performances we wish to predict, where B is the maximum age considered in the model.

The joint distribution of all season's best performances for athlete i can be partitioned like so:

$$\begin{bmatrix} y_{i,\text{obs}} \\ y_{i,\text{pred}} \end{bmatrix} \sim \text{N}\left(\begin{bmatrix} \mu_{\text{obs}} \\ \mu_{\text{pred}} \end{bmatrix}, \begin{bmatrix} \Sigma_{\text{obs,obs}} & \Sigma_{\text{pred,obs}} \\ \Sigma_{\text{obs,pred}} & \Sigma_{\text{pred,pred}} \end{bmatrix} \right) \tag{3.10}$$

where $y_{i,\text{obs}}$ are the observed season's best performances for athlete i, $y_{i,\text{pred}}$ are the unobserved performances that are to be predicted, $\mu_{\text{obs}}, \mu_{\text{pred}}$ are the mean vectors from the population polynomial, $\Sigma_{\text{obs,obs}}, \Sigma_{\text{pred,pred}}$ are the sub-matrices of the population covariance for observed and predicted ages respectively for athlete i , $\Sigma_{\text{pred,obs}}, \Sigma_{\text{obs,pred}}$ are the cross-covariance blocks.

Given the observed performances, the conditional predictive distribution for the unobserved performances is also a multivariate normal and takes the form:

$$y_{i,\text{pred}} \mid y_{i,\text{obs}} \sim \text{N}\left(\mu_{\text{pred}} + \Sigma_{\text{pred,obs}}\Sigma_{\text{obs,obs}}^{-1}(y_{i,\text{obs}} - \mu_{\text{obs}}),\ \Sigma_{\text{pred,pred}} - \Sigma_{\text{pred,obs}}\Sigma_{\text{obs,obs}}^{-1}\Sigma_{\text{obs,pred}}\right) \tag{3.11}$$

The conditional mean and covariance representations come from standard results of the conditional multivariate normal distribution [22]. This formulation ensures that predictions for future ages are informed by both the general age trajectory and the specific athlete's observed performances, with the uncertainty represented by the covariance matrix. This is different to future models, such as combined effects and random walks, as the individual deviations are captured within those models, however in this case those observed individual performances are what's considered in this conditional predictive distribution.

To implement this, the posterior samples of the model parameters, mean polynomial coefficients and covariance matrix, were first obtained. For athlete i the mean vector of predicted performances across all ages would be:

$$\boldsymbol{\mu}_i = f(a)$$

And so for each posterior MCMC sample s, the mean predicted performance at each age was calculated:

$$\boldsymbol{\mu_i}^{(s)} = X_{\text{poly}}\boldsymbol{\beta}^{(s)}$$

where X_{poly} is the matrix of polynomial basis functions evaluated at all ages and centred at at a central age. $\boldsymbol{\beta}^{(s)}$ are the sampled vector of the fixed effect polynomial coefficients, β_k. This gives the predicted mean vector for all ages in the age range considered, and are partitioned to ages that are observed for athlete i, to get μ_{obs} and unobserved future ages to get μ_{pred}.

Then for each posterior draw, the conditional mean and covariance of the unobserved ages were calculated using equation 3.11. Samples were then drawn from this conditional distribution in R, which allowed us to construct median predicted performance curves for each athlete and quantify uncertainty with credible intervals at each future age.

For each posterior sample s, the predicted vector of future observations for athlete i is found by:

$$y_{i,\text{pred}}^{(s)} \sim N(\mu_{\text{cond}}^{(s)}, \Sigma_{\text{cond}}^{(s)})$$

where the conditional mean and covariance are computed as in equation 3.11, using the sampled parameters. A typical visualization of this can be seen on the bottom row of plots in Figure 7. This provides an intuitive view of the model's fit and its uncertainty about future performance.

This fixed effect model is simple and interpretable, capturing the average age-performance relationship. However, it does not directly account for persistent, athlete-specific deviations from the population mean. It cannot distinguish athletes who consistently outperform or underperform their peers at each age, nor can it capture individual patterns of progression. Although this behaviour is implied in the conditional predictive distribution. This motivates the use of a more flexible mixed effect model, which is discussed in the following Section 3.2.5.

3.2.5 Combined Effects Model

We combine the fixed effect and random effects by formally modelling the season's best, $y_{i,a}$, for athlete i at age a by the following:

$$y_{i,a} = f(a) + \gamma_{i,a} + \epsilon_{i,a}$$

- $y_{i,a}$ - season's best result for athlete i at age a
- $f(a)$ - fixed effect, population level performance
- $\gamma_{i,a}$ - random effect, which is the excess performance at each age (difference between athlete's performance level and population average performance level)

The residual errors are independent and normally distributed with zero mean and standard deviation σ_y. Only observed $y_{i,a}$ values are included in the likelihood:

$$y_{i,a} \sim N(f(a) + \gamma_{i,a}, \sigma_y^2), \qquad i = 1, \ldots, N,\ a = A, \ldots, B$$

$$\sigma_y \sim \text{Half-Cauchy}(0,\ 2.5)$$

To handle missing values, we specify the likelihood only over observed entries using the indexing method specified in Section 3.2.1. $f(a)$ takes the same form described in equation 3.7 and let $\boldsymbol{\gamma_i}$ be the athlete-specific random effect vector for athlete i as described in 3.5.

For each athlete i, we define obs_count$[i]$ as the number of available observations and age_idx$[i, :]$ as the indices of ages for which performance data were available. The likelihood for each athlete was computed only over these observed entries, explicitly ignoring any missing data points. By structuring the likelihood and prior distributions this way, we ensured a robust, computationally efficient approach to modelling the available season's best data.

The priors for the $\gamma_{i,a}$'s follow the same distributions as in equation 3.9 and for the fixed effect, $f(a)$, the priors for β_k take the same form in equation 3.8.

The DAG for this model can be shown in Figure 5.

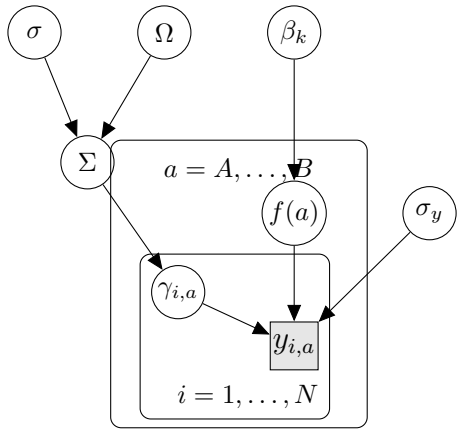

Figure 5: DAG of Combined Effects hierarchical model

As motivation for this particular model, we aim to balance the fixed effects and random effects. To build intuition, we first looked at the polynomial fit of the population level age effect, where least squares was used to get a sense of this fixed effect part (described in Section 3.2.4). Conceptually, the random effects can then be thought of as capturing the individual deviations from this mean age effect, playing a role similar to residuals.

Presenting this formally, suppose we let $\hat{f}(a)$ be the least squares fit of the fixed effect. Then we could write

$$y_{i,a} = \hat{f}(a) \ + \gamma_{i,a} + \epsilon_{i,a}$$

with residuals written as:

$$r_{i,a} = y_{i,a} - \hat{f}(a)$$

$$\approx \gamma_{i,a} + \epsilon_{i,a}$$

Averaging over repeated samples, these residuals would satisfy:

$$\bar{r}_{i,a} \ \approx \ \bar{\gamma}_{i,a}, \quad \text{since } \epsilon_{i,a} \sim N(0, \sigma_y^2)$$

where $\bar{\gamma}_{i,a}$ are the posterior mean gammas. This framing shows how the $\gamma_{i,a}$ terms can be interpreted as individual deviations from the population curve, being the residuals.

3.2.5.1 Predictive Distribution

A key advantage of this combined effects model is its ability to generate individualised predictions for each athlete, accounting for both the population trend and the athlete's own unique performance trajectory which is captured by their random effect.

Given the posterior samples from the model, that is, samples of the fixed effect coefficients and the random effects for each athlete, we can construct the posterior predictive distribution for any age, observed or unobserved. This distribution incorporates both model uncertainty (via the spread in posterior samples) and the athlete's past performances, which is through the estimated random-effects trajectory.

For athlete i let the vector of predicted performances across all ages be:

$$\boldsymbol{\mu}_i = f(a) + \boldsymbol{\gamma}_i$$

For each posterior MCMC sample s, we can calculate the mean predicted performance at each age:

$$\boldsymbol{\mu_i}^{(s)} = X_{\text{poly}}\boldsymbol{\beta}^{(s)} + \boldsymbol{\gamma_i}^{(s)}$$

where X_{poly} is the matrix of polynomial basis functions evaluated at all ages and centred at at a central age, $\boldsymbol{\beta}^{(s)}$ are the sampled vector of fixed effect coefficients, β_k, and $\boldsymbol{\gamma_i}^{(s)}$ is the sampled random effect vector for athlete i.

This gives the predicted mean vector for all ages in the age range considered, whether those ages are observed or not. We can also sample the posterior residual standard deviation, $\sigma_y^{(s)}$. Repeating this across all posterior draws provides the empirical posterior predictive distribution for an athlete at a single age, which is Normal:

$$y_{i,a}^{(s)} \sim N(\mu_{i,a}^{(s)}, \sigma_y^{(s)})$$

This sampling-based approach provides the basis for constructing median predicted trajectories and credible intervals.

Alternatively, the model can also be expressed at the joint level, where the joint distribution can be partitioned in the same way as in equation 3.10. Since the sum of independent Normal terms is also Normal, the vector of performances follows a multivariate normal distribution of the form:

$$\boldsymbol{y}_i \sim N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma} + \sigma_y^2\boldsymbol{I})$$

Predictions for unobserved ages given observed results can be obtained using the standard conditional multivariate Normal formula as done in equation 3.11.

For observed ages, predictions can be compared directly to data. For unobserved ages, these samples form the full posterior predictive distribution for that athlete's season's best. Across all posterior samples, we can summarise the predictive distribution at each age e.g., by reporting the median and 95% credible intervals.

This approach gives personalised predictions for each athlete. The predictive distribution naturally reflects greater uncertainty at ages where an athlete has no data, but shrinks toward their estimated random effect at ages with observations. This also allows us to answer the motivating questions (described in Section 1.2) in a probabilistic way by summarising across posterior samples.

3.2.6 1st Order Random Walk Model

In addition to the individual random effects, a 1st order random walk model can be implemented for the athlete-specific deviations. The implementation of a random walk model was decided, as initial results from previous models showed weak correlations between ages, for example in Table 3, a stronger correlation effect was expected.

This approach retains the population-level polynomial, but we replace the multivariate normal prior on the random effects with a random walk prior. Here, the excess performance of an

athlete (over the population average) progresses gradually with age, where the deviation of each year is informed by the previous year.

The initial setup is the same as in Section 3.2.5 where:

$$y_{i,a} = f(a) + \gamma_{i,a} + \epsilon_{i,a}$$

However this time instead of drawing $\boldsymbol{\gamma_i} = (\gamma_{i,A}, \ldots, \gamma_{i,B})$ from a multivariate normal distribution with a fully estimated covariance matrix, we use a first-order random walk:

$$\begin{aligned} \gamma_{i,A} &\sim N(0, \sigma^2), \\ \gamma_{i,a} \mid \gamma_{i,a-1} &\sim N(\gamma_{i,a-1}, \tau^2), \quad a = A+1, \ldots, B, \\ \sigma, \tau &\sim \text{Half-Cauchy}(0, 2.5). \end{aligned} \tag{3.12}$$

σ controls the initial variability of the youngest age and τ controls variability of year-to-year changes in the random effect. Small τ correspond to smoother, more stable deviations, while larger values allow more rapid year-to-year changes.

This random walk prior has the capability of reflecting realistic physiological and training effects. Unlike the full multivariate normal prior in the combined effects model, the random walk formulation avoids estimating a large $J \times J$ covariance matrix, making it more computationally stable for larger J. The parameters, σ and τ have direct interpretations being initial variability and year-to-year performance volatility respectively.

A Gaussian assumption was used for the random walk increments $(\gamma_{i,a} - \gamma_{i,a-1})$ but it is possible to replace these with a Student-t distribution to allow for heavier tails, accommodating for the possibility of occasional large jumps in performance between seasons, for example due to injury, breakthrough improvements, or major changes in training.

As in Section 3.2.5 only observed $y_{i,a}$ values are included in the likelihood:

$$\begin{aligned} y_{i,a} &\sim N\big(f(a) + \gamma_{i,a},\, \sigma_y^2\big), \quad i = 1, \ldots, N,\ a = A, \ldots, B, \\ \sigma_y &\sim \text{Half-Cauchy}(0, 2.5). \end{aligned} \tag{3.13}$$

3.2.7 2nd Order Random Walk Model

For smoother athlete-specific trajectories, than the first-order random walk, we can also consider a second-order random walk model. As done with the first order differences, a Gaussian prior is also placed on the second differences:

$$\begin{aligned} \gamma_{i,A} &\sim N(0, \sigma_1^2), \\ \gamma_{i,A+1} &\sim N(0, \sigma_2^2), \\ \gamma_{i,a} \mid \gamma_{i,a-1}, \gamma_{i,a-2} &\sim N\big(2\gamma_{i,a-1} - \gamma_{i,a-2},\, \tau^2\big), \quad a = A+2, \ldots, B, \\ \sigma_1,\ \sigma_2,\ \tau &\sim \text{Half-Cauchy}(0, 2.5). \end{aligned} \tag{3.14}$$

In the first order random walk you would get the strongest dependence between the immediate neighbours of the random effects i.e $\gamma_{i,a-1}$ and $\gamma_{i,a}$. However in this model, correlations are likely to be weaker the larger the differences in ages get. This may be a reasonable assumption if we believe age deviations for an athlete are short memory. But it may be the case that in

athlete careers, we may expect longer age range coherence. For example a strong season's best performance at age 21 could correlate well with performances at age 22 and 23, such as athlete 2 in Figure 7.

The key component of modelling athlete-specific deviations with a second order random walk is that it preserves stronger correlations across multiple ages, not just between consecutive years [23]. As a result, if an athlete's trajectory is trending upward at age, a, the second order random walk makes it more likely that a trend will persist for several years. The likelihood takes the same form in as in equation 3.13.

4 Results

4.1 Model Evaluation Metrics

To evaluate the predictive performance of the proposed models, we use an out-of-sample validation set, where the final observed season's best performance for each athlete is masked during model fitting. This is an example of leave-one-out cross-validation (LOO-CV) which is applied per athlete, and each athlete's final year season's best result forms their hold-out set. LOO-CV is widely recommended in Bayesian predictive modelling [24] because it uses all available data for both training and validation, providing low-bias estimates of predictive performance. This setup simulates the real world scenario of forecasting unobserved future performance based on the historical results of an athlete.

We evaluate the models using a few metrics. Root Mean Squared Error (RMSE), measures the average magnitude of the prediction error [25] and is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\bar{y}_{i,a_i} - y_{i,a_i}^{\text{true}})^2}$$

where $\bar{y}_{i,a_i}$ is the posterior mean prediction obtained by averaging over S posterior predictive draws: $\bar{y}_{i,a_i} = \frac{1}{S}\sum_{s=1}^{S} y_{i,a_i}^{(s)}$ and y_{i,a_i}^{true} is the masked, held-out season's best for athlete i at their final observed age a_i.

The Log Predictive Score (LPS) evaluates the quality of the full predictive distribution. Following [26] [27], the LPS for K-fold validation is given by:

$$\text{LPS} = -\frac{1}{N}\sum_{i=1}^{N}\log\left[\frac{1}{S}\sum_{s=1}^{S} p(y_{i,a_i}^{\text{true}} \mid \boldsymbol{\theta}^{(s)})\right]$$

where $p(y_{i,a_i}^{\text{true}} \mid \boldsymbol{\theta}^{(s)})$ is the predictive density under the sth posterior draw of parameters $\boldsymbol{\theta}^{(s)}$. This will be defined differently for each model:

- Random effect: $\boldsymbol{\theta}^{(s)} = (\gamma_{i,a}^{(s)}, \sigma_y^{(s)}, \boldsymbol{\sigma}^{(s)}, \boldsymbol{\Omega}^{(s)}, \boldsymbol{\Sigma}^{(s)})$
- Fixed effect: $\boldsymbol{\theta}^{(s)} = (\beta^{(s)}, \boldsymbol{\sigma}^{(s)}, \boldsymbol{\Omega}^{(s)}, \boldsymbol{\Sigma}^{(s)})$
- Combined effects: $\boldsymbol{\theta}^{(s)} = (\beta^{(s)}, \gamma_{i,a}^{(s)}, \boldsymbol{\sigma}^{(s)}, \boldsymbol{\Omega}^{(s)}, \boldsymbol{\Sigma}^{(s)}, \sigma_y^{(s)})$

- 1st order Random walk: $\boldsymbol{\theta}^{(s)} = (\beta^{(s)}, \gamma_{i,a}^{(s)}, \sigma^{(s)}, \tau^{(s)}, \sigma_y^{(s)})$
- 2nd order Random walk: $\boldsymbol{\theta}^{(s)} = (\beta^{(s)}, \gamma_{i,a}^{(s)}, \sigma_1^{(s)}, \sigma_2^{(s)}, \tau^{(s)}, \sigma_y^{(s)})$

Lower values of LPS indicate better predictive performance.

These two particular metrics are especially useful in measuring model performance since RMSE measures accuracy of the mean prediction and LPS measures how well calibrated the predictive distributions are. Both are important for our application, where we require both precise and well-calibrated uncertainty estimates.

4.2 Out-of-Sample Prediction Results

Model	**Male**		**Female**	
	RMSE	**LPS**	**RMSE**	**LPS**
Random Effects	0.1515987	-0.4440104	0.2125748	-0.1201231
Fixed Effects	0.1523213	-0.4576526	0.2094711	-0.1414431
Combined Effects	0.1569125	-0.09483849	0.2109678	-0.04899084
Random Walk 1st Order	0.1572718	-0.41471	0.2147766	-0.1018274
Random Walk 2nd Order	0.16787	-0.3707398	0.234912	-0.03747368

Table 1: Out-of-sample RMSE and log predictive scores by gender

Table 1 reports the out-of-sample root mean squared error (RMSE) and log predictive score (LPS) for each of the five models considered, evaluated separately for male and female athletes.

For male athletes, the random effects model and the fixed effects model achieved the smallest RMSE and LPS values, indicating better forecasting. The Combined Effects and Random Walk models showed higher RMSEs and worse LPS values (closer to -0.09 and -0.37), suggesting that they assign lower predictive probability to unseen outcomes.

For female athletes, the Fixed Effects model again provided the best LPS of -0.141 and the lowest RMSE of 0.2095. The Random Effects, Combined Effects and first-order Random Walk models performed slightly worse than this. The second-order Random Walk showed the weakest performance, with both the highest RMSE of 0.235 and least negative LPS of -0.037. Metric values differ between male and female due to different sizes in the leave-one-out cross-validation test set as for male, there are 672 performances and for female there are 518 performances.

It is important to note that RMSE values should be interpreted relative to the estimated residual standard deviation. For the random and combined effects model, posterior draws suggest σ_y is around 0.02–0.04 seconds. These small values may be explained due to most of the variability being captured by the athlete-specific random effects. The random walk models have slightly larger residual standard deviations ($\sigma_y \approx 0.10 - 0.15$), reflecting that year-to-year fluctuations are only partly explained by the walk. The fixed effects model has no explicit residual term, but its covariance matrix implies marginal standard deviations (σ_j's). These are around 0.18 – 0.22 seconds for males and 0.21 - 0.29 for females, closely matching the out-of-sample RMSE values (0.15–0.23). This alignment suggests that the fixed effects model captures most of the predictable structure in the data, which further supports the strength of the fixed effects model.

Overall, the out of sample results indicate that the simpler Fixed and Random Effects models generalise best in out-of-sample prediction as opposed to the more complex models. This could

indicate that the complex models tend to overfit. Between the Fixed and Random Effects, the Fixed Effects model shows the strongest and most consistent performance, so this is the model that will be used in answering predictive questions.

4.3 Results Analysis

Figure 6 shows the posterior average population performance trajectory, from the fixed effects model, as a function of age, a. For ages between 17 and 30, the plots suggest that age of peak performance for males is around 24 years and for females it is around 26 years and this is supported by [11] which suggested that peak performance occurs two years earlier for males. For both males and females there is a rapid improvement between ages 17 and 21, which is also supported by [12] as this saw a similar effect for 100 metre and 200 metre freestyle swimming. The posterior estimates for the coefficients of this quartic fixed effect polynomial, $f(a)$, for the modelling of male athlete are displayed in Table 2, together with standard deviations, effective sample sizes, $\hat{R}$ diagnostics and posterior quantile summaries. Diagnostics indicate a satisfactory mixing with $\hat{R} \approx 1$ and large effective sample sizes. From the interpretations described in equation 3.7, the posterior estimate of β_0 suggest that the average season's best at 23.5 years is approximately 10.36.

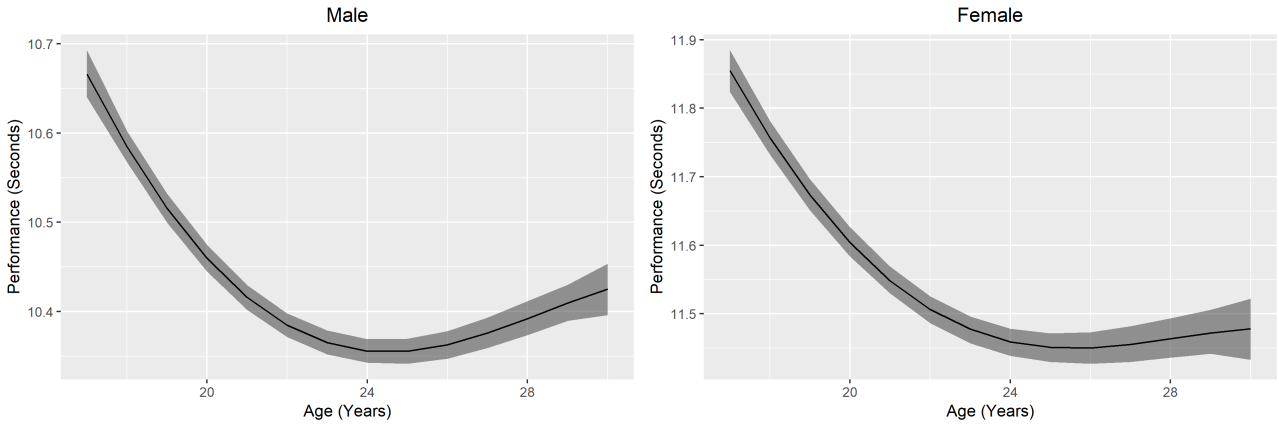

Figure 6: Average population performance trajectory, $f(a)$, for both male and female athletes in 100 metre sprint. The trajectories are shown as posterior median (black line) and 95% credible interval (grey shading)

Coefficient	Rhat	n_eff	mean	sd	2.5%	50%	97.5%
β_0	0.999549	870	10.359423	0.006859	10.346350	10.359256	10.372955
β_1	0.999745	2060	-0.009267	0.001886	-0.012992	-0.009298	-0.005588
β_2	1.001055	1170	0.004933	0.000644	0.003662	0.004930	0.006176
β_3	0.999339	1167	-0.000220	0.000059	-0.000335	-0.000218	-0.000108
β_4	1.000759	1219	-0.000012	0.000016	-0.000043	-0.000012	0.000018

Table 2: Posterior summary for coefficients β_k used in $f(a)$ for male athlete modelling

Figure 7 shows the average population performance trajectory, overlaid with observed performances (top row) and the conditional predictive trajectories (bottom row) of three male 100 metre athletes. The top row of the figure demonstrates that individual athlete performance can differ significantly from the average population and just relying on the mean effect is not good enough to predict future performance. This highlights the necessity of the conditional

predictive distribution which is informed by the previous observations as well as the average general age trajectory. This can be seen in the bottom row of the figure where each singular future age, for a particular athlete, uses the conditional predictive formula in equation 3.11 and then averaged over all samples to get the posterior conditional median trajectory. For example, for athlete 1, to get the conditional predictive at age 19, we are conditioning only on age 18 (which is the only observation before 19) to get a prediction for age 19. Then for age 20 we are conditioning on the observations from age 18 and 19 to get the prediction for age 20 and so on. This gives us the trajectories shown in the bottom row of the figure. The shaded credible interval is simply the 2.5% and 97.5% quantiles of the predictive draws, calculated across all the posterior samples.

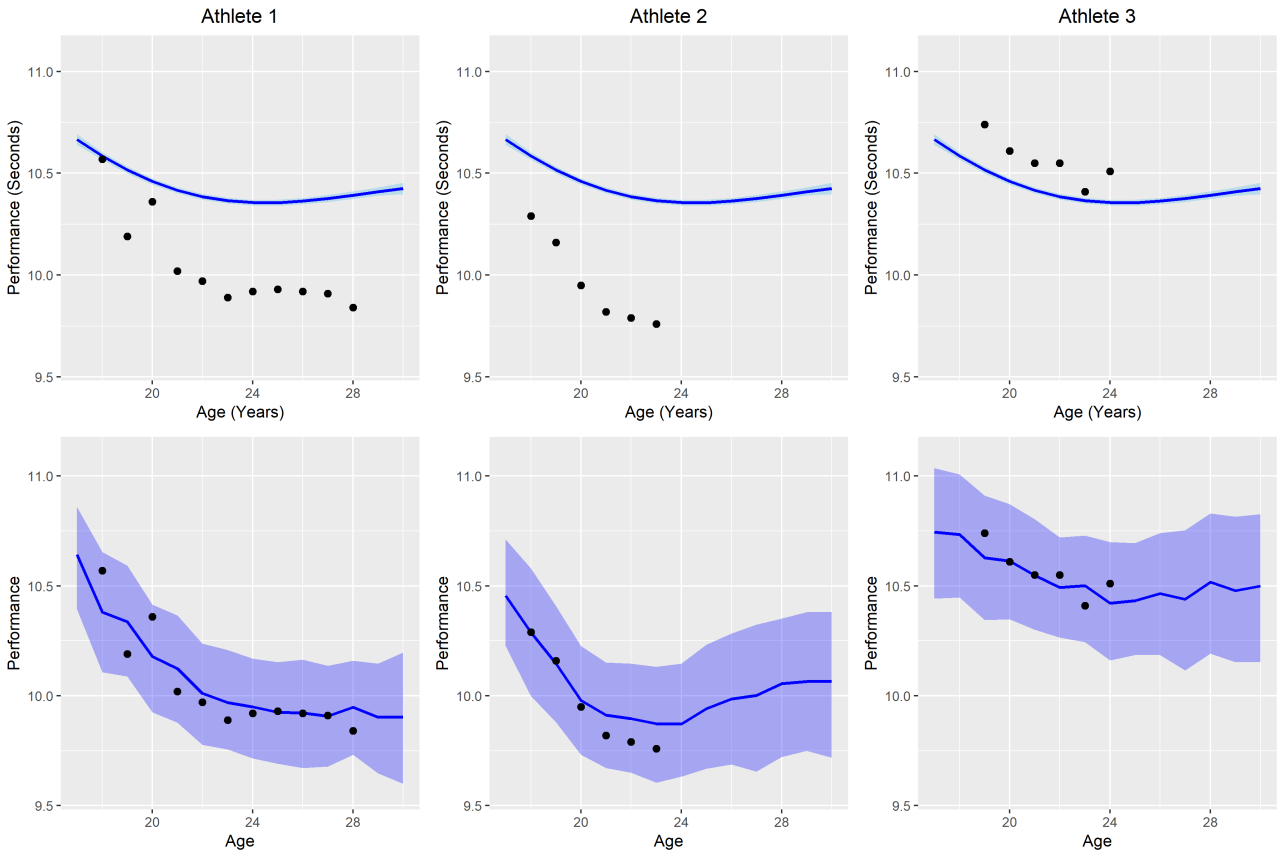

Figure 7: Average population performance trajectories (top row) and trajectories from conditional predictive distribution (bottom row) for three male 100 metre athletes. The conditional predictive trajectories are shown as posterior median (solid blue line) and 95% credible interval (blue shading) with observed performances (black dots).

The conditional predictive median aligns closely with the observed performances for all athletes, for the most part, and all observations fall within the predictive intervals. However, we see that these credible intervals are quite wide, highlighting the high uncertainty in the predictions, even with historical observed performances. The predictions for unobserved performances can explicitly be seen for athletes 2 and 3. What is particularly interesting for athlete 2, is that his observed performances all follow a consistent improvement, but the model forecasts a slight decline in performance from age 24 onward, which seems counterintuitive. This reflects the influence of the average population age trajectory (shown by the plot directly above it) outweighing the immediate past results. This could be attributed to the lack of correlation between ages. If we examine the posterior correlation matrix, shown in Table 3, adjacent ages exhibit reasonably high correlations around 0.7 but as the age gap increases the correlation decreases quite rapidly. While the correlation structure does allow for some borrowing of information

across ages, it is not strong enough to capture larger age ranges in individual trajectories. This limitation motivated the random walk models in Section 3.2.6 and 3.2.7, but from the out of sample results in Table 1, these models did not perform as well as the fixed effects model.

Age	17	18	19	20	21	22	23	24	25	26	27	28	29	30
17	1.000	0.719	0.453	0.395	0.329	0.244	0.332	0.339	0.335	0.291	0.180	0.258	-0.019	0.322
18	0.719	1.000	0.607	0.526	0.424	0.414	0.361	0.383	0.268	0.244	0.142	0.339	0.152	0.308
19	0.453	0.607	1.000	0.614	0.551	0.485	0.417	0.386	0.285	0.336	0.171	0.406	0.204	0.341
20	0.395	0.526	0.614	1.000	0.702	0.576	0.543	0.485	0.421	0.391	0.362	0.361	0.252	0.243
21	0.329	0.424	0.551	0.702	1.000	0.648	0.627	0.532	0.464	0.452	0.369	0.407	0.380	0.372
22	0.244	0.414	0.485	0.576	0.648	1.000	0.693	0.625	0.558	0.482	0.445	0.475	0.432	0.395
23	0.332	0.361	0.417	0.543	0.627	0.693	1.000	0.711	0.634	0.552	0.528	0.467	0.502	0.551
24	0.339	0.383	0.386	0.485	0.532	0.625	0.711	1.000	0.681	0.657	0.605	0.556	0.541	0.536
25	0.335	0.268	0.285	0.421	0.464	0.558	0.634	0.681	1.000	0.663	0.612	0.641	0.538	0.570
26	0.291	0.244	0.336	0.391	0.452	0.482	0.552	0.657	0.663	1.000	0.663	0.641	0.613	0.521
27	0.180	0.142	0.171	0.362	0.369	0.445	0.528	0.605	0.612	0.663	1.000	0.743	0.658	0.613
28	0.258	0.339	0.406	0.361	0.407	0.475	0.467	0.556	0.641	0.641	0.743	1.000	0.727	0.676
29	-0.019	0.152	0.204	0.252	0.380	0.432	0.502	0.541	0.538	0.613	0.658	0.727	1.000	0.716
30	0.322	0.308	0.341	0.243	0.372	0.395	0.551	0.536	0.570	0.521	0.613	0.676	0.716	1.000

Table 3: Posterior mean correlation matrix $\mathbf{\Omega}$ (ages 17-30)

4.4 Predictive Questions Results

Having identified the Fixed Effects model as the best-performing approach, based on the out of sample inference in Section 4.2, we now use its posterior predictive distribution to address the three applied research questions relating to talent identification and forecasting elite performance discussed in 3.2.2. These questions reflect real world decision making scenarios in athletics, such as when to allocate funding, how to evaluate athletes relative to peers, and how to anticipate championship finalists.

4.4.1 When age to decide funding?

Using the posterior predictive draws from the fixed effect model, as discussed in section 3.2.4.1, we simulate the distribution of season's best performances at a particular age a, given all their performances up to age A^*, where $a > A^*$. The probability in equation 3.1 is what needs to be estimated.

We chose $T = 9.9$ for male athletes since a sub-10 second time is considered elite level in modern times [28]. We can calculate this quantity for multiple different future ages, a, and we choose the age that yields the largest probability of reaching elite status as the age at which funding should be decided. This is done like this since this is the age we are most confident for elite performance. This probabilistic framework is consistent with approaches to talent identification that highlight the importance of forecasting uncertainty rather than relying on deterministic thresholds [29] [30].

These quantities are calculated using the MCMC samples. Since we have the posterior MCMC samples of future performances, y, for each athlete and age, we estimate $P(y_{i,a} < T \mid y_{i,1}, \cdots, y_{i,A^*})$ using the following procedure:

For each posterior sample, s (where $s = 1, \ldots, S$), and for a specified age, a, calculate:

$$\hat{P}(y_{i,a} < T \mid y_{i,1}, \ldots, y_{i,A^*}) = \frac{1}{S} \sum_{s=1}^{S} \mathbf{1}\Big(y_{i,a}^{(s)} < T\Big)$$

Here we are essentially counting how many times $y_{i,a}^{(s)} < 9.9$ and then averaging over all the samples, S. This gives us an estimate for the probability in equation 3.1 using MCMC samples.

Table 4 shows the predictive probabilities of achieving elite status for athletes aged 24 or younger at the time of the dataset, with probabilities shown across their future subsequent ages. The cut-off of 24 was chosen arbitrarily, simply to provide an illustration of how the model outputs can be interpreted in practice. The table is ordered by probability, so only the highest ranked athletes are displayed. For instance, Christian Coleman is assigned the greatest probability of reaching elite status at age 25 and so sponsors may decide funding to start as soon as possible at the age of 25 for him. However, Christian Coleman is quite a prominent athlete, being the 2019 world champion for 100 m, so such results are unsurprising. The value of this approach becomes clearer for less well known athletes, such as Leshon Collins, where age 27 is the age that the model is most confident on for elite performance. This could provide useful guidance for talent identification and funding decisions.

Competitor	**Age**	**Prob. Elite**
Christian COLEMAN	25	0.4107
Christian COLEMAN	26	0.3147
Christian COLEMAN	27	0.2987
Christian COLEMAN	28	0.1953
Christian COLEMAN	29	0.1840
Christian COLEMAN	30	0.1780
Leshon COLLINS	27	0.1033
Beejay LEE	27	0.0833
Leshon COLLINS	26	0.0693
Beejay LEE	25	0.0640
Beejay LEE	26	0.0600
Jackson WEBB	30	0.0587
Marvin SCHULTE	27	0.0580
Filippo TORTU	25	0.0567
Markesh WOODSON	27	0.0560
Demi OMOLE	26	0.0540
Demi OMOLE	28	0.0520
Demi OMOLE	30	0.0520
Filippo TORTU	26	0.0507
Cravont CHARLESTON	26	0.0500
Filippo TORTU	27	0.0453
Chad WALKER	27	0.0447
Jackson WEBB	25	0.0440
Leshon COLLINS	25	0.0440
Filippo TORTU	28	0.0427

Table 4: Predicted elite performance probabilities for leading male athletes aged below 24

4.4.2 How well is an athlete compared to peers of the same age?

We estimate equation 3.2 by finding $y_{i,a}^{(s)}$ and $y_{k,a}^{(s)}$ for each posterior sample s. Then the number of occurrences at which $y_{i,a}^{(s)}$ is less than $y_{k,a}^{(s)}$ is counted and then averaged. This can be shown mathematically as follows:

$$\hat{P}(y_{i,a} < y_{k,a} \,|\, \boldsymbol{y}) = \frac{1}{S} \sum_{s=1}^{S} \mathbf{1}\Big(y_{i,a}^{(s)} < y_{k,a}^{(s)}\Big) \tag{4.1}$$

To estimate the extension of finding the probability that a specific athlete i is faster than all others at a fixed age a, as in equation 3.3, the following procedure is used:

For each posterior sample, s, find:

$$\begin{pmatrix} y_{1,a}^{(s)} \\ y_{2,a}^{(s)} \\ \vdots \\ y_{N,a}^{(s)} \end{pmatrix}$$

and do this for each athlete 1 through to N. This gives the following matrix:

$$\begin{pmatrix} y_{1,a}^{(1)} & y_{1,a}^{(2)} & \cdots & y_{1,a}^{(S)} \\ y_{2,a}^{(1)} & y_{2,a}^{(2)} & \cdots & y_{2,a}^{(S)} \\ \vdots & \vdots & \ddots & \vdots \\ y_{N,a}^{(1)} & y_{N,a}^{(2)} & \cdots & y_{N,a}^{(S)} \end{pmatrix}$$

Then within each of these posterior samples, find the minimum time and the athlete associated with this minimum time, to get, $k^{*(s)}$:

$$k^{*(s)} \in \underset{k \in \{1,\ldots,n\}}{\operatorname{argmin}} \big(y_{k,a}^{(s)}\big)$$

where $k^{*(s)}$ is the fastest athlete in sample s.

Repeat this for each s, 1 through to S to obtain the vector:

$$(k^{*(1)}, k^{*(2)} \cdots, k^{*(S)})$$

Then estimate the probability using:

$$\hat{P}(y_{i,a} < y_{k,a} \;\; \forall\, k \neq i \,|\, \boldsymbol{y}) = \frac{1}{S} \sum_{s=1}^{S} \mathbf{1}\big(k^{*(s)} = i\big) \tag{4.2}$$

As an example, we can compare two athletes considered in the dataset. Let athlete i correspond to Filippo Tortu and let athlete k correspond to Jack Hale. Both are talented 100 m athletes who are the same age, of 23 years old, and have varying performance levels historically. A probability of Tortu being faster than Hale at a future age a can be estimated using the

approach in 4.4.2. Let a be 27 years and then apply the formula for the estimated probability in 4.1. Doing this calculation we obtain $\hat{P}(y_{i,a} < y_{k,a} \mid \boldsymbol{y}) = 0.572$. This is the models estimate of Filippo Tortu being faster than Jack Hale when they are at the age of 27. In fact the age of 27 corresponds to the age both athletes are for the 2025 season and from the latest data, Filippo Tortu has a season's best of 10.19 and Jack Hale has a season's best of 10.22 [31] in 2025. The model's probability estimate aligns with the observed ordering of performances. Such pairwise comparisons illustrate how the model can generate binary outcome probabilities that are interpretable.

We can also use the extension where the same methodology can be applied to all athletes to develop a ranking for a future age. Using the estimated probability formula in 4.2, for all current 23 year old athletes, we get the ranking shown in Table 5. Here a is set to 28 years old.

Competitor	**Prob. Best**
Christian Coleman	0.291
Demi Omole	0.093
Cravont Charleston	0.081
Filippo Tortu	0.075
Aaron Ernest	0.053
Felipe Bardi	0.040
Kenzo Cotton	0.032
Arthur Delaney	0.013
David Winters	0.023
Jack Hale	0.023
Maxwell Willis	0.022
Nic Bowens	0.017
Jeremy Hall	0.016
Amir James	0.013
Cliff Resias	0.012

Table 5: Predicted probabilities of being the best performer at age 28 among male 23 year old 100 m athletes

4.4.3 Who's likely to reach a championship final?

Since our dataset consists of performances up to August 2021, described in Section 3.1, the 2022 World Championships serves as a good reference competition to predict for.

The relevant probability in equation 3.4 can be estimated by ranking the posterior predictive draws of season's best performances across all athletes in the dataset and estimating the probability that athlete i is among the eight fastest. Repeating this for each athlete yields predicted finalist probabilities and an implied ranking of the athletes. Showing this algorithmically, for each posterior sample, s, rank all athletes, at age a_i^*, 1 to N to find:

$$\begin{pmatrix} y^{(s)}_{1,a_1^*} \\ y^{(s)}_{2,a_2^*} \\ \vdots \\ y^{(s)}_{N,a_N^*} \end{pmatrix}$$

Then for each athlete i and each sample, s, define a latent variable:

$$T_i^{(s)} = \begin{cases} 1, & \text{if rank}_i \leqslant 8 \\ 0, & \text{otherwise} \end{cases}$$

Using this, the following probability can be estimated:

$$\begin{aligned} \hat{P}(\text{athlete } i \text{ in top 8 at age } a_i^* \mid \boldsymbol{y}) &= \frac{1}{S}\sum_{s=1}^{S} T_i^{(s)} \\ &= \frac{1}{S}\sum_{s=1}^{S} \mathbf{1}\Big(\text{rank}_i^{(s)} \leqslant 8\Big) \end{aligned}$$

In this approach it is important to note that predictions are conditional on the dataset. Some actual championship finalists may not appear, either because they did not meet our inclusion criteria (minimum number of season's bests, age restrictions, or elite performance cut-off) or were absent from the original dataset. Therefore, our estimates should be interpreted as relative probabilities within the athletes considered in this study rather than predictions of the exact championship field.

Table 6 shows this prediction for current male athletes and their projection for the 2022 season, in particular the 2022 World Athletics Championships. Studies have shown that athletes tend to reach their season's best performance during the time of a major competition within that year [12], for e.g the World Championships or Olympics. So projecting to the World Championship finals can be a reasonable assumption. The table shows the athletes projected to be in the top 8 and their actual performance in this particular world championship.

Competitor	**Nat**	**Prob. Final**	**Stage Reached**
Christian Coleman	USA	0.783	Final
Akani Simbine	RSA	0.735	Final
André De Grasse	CAN	0.559	Semi-Final
Lamont Marcell Jacobs	ITA	0.385	Semi-Final
Jimmy Vicaut	FRA	0.356	DNC
Cravon Gillespie	USA	0.222	DNC
Filippo Tortu	ITA	0.190	DNC
Marvin Bracy	USA	0.188	Final

Table 6: Top 8 predicted probabilities of reaching 2022 final with actual competition outcome for male athletes

Despite the limitation of not including every world class athletes in the dataset, there were some projections that aligned closely with observed outcomes in the 2022 World Championships. Christian Coleman and Akani Simbine were assigned the highest probabilities of reaching the final (0.783 and 0.735 respectively) and both ultimately reached the final. Marvin Bracy was assigned a low probability (0.188) but did in fact progress to the final and was ultimately placed in second [32]. However there were cases for which the model did not predict accurately, André De Grasse and Lamont Marcell Jacobs had moderate probabilities but progressed only to the semi-final stage. In Jacobs' case, this was due to injury withdrawal [33], which made final

qualification impossible regardless of prior ability. For the remaining athletes listed in Table 6, non-qualification or 'did not compete' (DNC) was primarily due to competition structure and selection rules. For example, Cravon Gillespie did not make the U.S. team, as each nation is limited to a maximum of three qualified athletes (or 4 in the special case for US) [34]. The U.S. qualifiers for 2022 were Fred Kerley, Marvin Bracy, Trayvon Bromell, and Christian Coleman, all of whom reached the final.

This highlights the difficulty in predicting future finalists, partly attributed to the added complexities of injuries, national selection processes and other contextual factors that fall outside the scope of performance-based prediction. But this also highlights a limitation on focusing on season's bests, as athletes are excluded from analysis if they don't have enough recorded performances.

5 Discussion

5.1 Strengths of the study

Overall, this study has demonstrated several strengths. By focusing on season's best performances we were effectively able to provide an interpretable measure of peak annual ability. This choice reduces the influence of injuries or suboptimal conditions, which are more prevalent in prior studies that use all competition results within a year [1]. Multiple modelling approaches were implemented and compared. This gave us an understanding of how different assumptions, like more correlations in random walk models, affect predictive performance. A further technical strength is the novel indexing method for handling missing data, which avoids unrealistic imputation and ensures inference is based only on observed results.

Another important strength is the use of predictive distributions, which allowed us to move beyond point estimates and quantify uncertainty. This methodological choice is what made it possible to frame and answer applied predictive questions, demonstrating its practical utility. Real predictive scenarios were considered that coaches, sponsors or athletes can benefit from. Examples include deciding when to fund future talent, benchmarking athletes relative to peers and projecting finalists for major championships, highlighting the broader potential of this modelling framework. Beyond these examples, the framework could also be extended to address additional questions, such as how long an athlete can maintain elite performance, which junior athletes are most likely to transition successfully to senior athletics and how current trajectories compare with historic medal finalists and many more. This highlights the versatility of the approach, showing that the same framework can be adapted to a wide range of applied questions in elite sport.

5.2 Limitations and extensions

The models developed in this study provide valuable insights into athlete performance trajectories, however there are several limitations that could lead to natural extensions for future work.

Firstly our dataset was reduced to each athlete's season's best rather than the full set of performances within a competitive year. This was chosen due to its simplicity and interpretability. Despite this, the approach discards within-season variation in performance which could provide richer information about form and progression. A natural extension would be to model all

observed performances within each season, potentially using hierarchical structures to separate within-season variability from long-term developmental trends [5].

The fixed effect polynomial model used in the final analysis is simple, interpretable, and consistent with prior literature on age performance curves [6]. However, it does not account for athlete-specific deviations from the population mean, so it cannot distinguish athletes who consistently outperform or underperform their peers at each age.

The error distribution in our models was assumed to be Gaussian for computational tractability but sprint times often show mild skewness and heavier tails. An alternative would be to model errors with skew-t or other heavy tailed distributions [35], [11]. Attempts to implement skew-t likelihoods in Stan proved numerically challenging, but this remains a promising extension as probabilistic programming frameworks advance [36]. There are also possibilities of exploring different multivariate distributions for the random effects, not just multivariate normal which was implemented in all the models in this study.

References

[1] David B Pyne, Cassie B Trewin, and William G Hopkins. Progression and variability of competitive performance of olympic swimmers. *Journal of sports sciences*, 22(7):613–620, 2004.

[2] Dennis-Peter Born, Eva Rüeger, C Martyn Beaven, and Michael Romann. Comparing cross-sectional and longitudinal tracking to establish percentile data and assess performance progression in swimmers. *Scientific Reports*, 12(1):10292, 2022.

[3] Sian V Allen, Tom J Vandenbogaerde, and William G Hopkins. Career performance trajectories of olympic swimmers: Benchmarks for talent development. *European journal of sport science*, 14(7):643–651, 2014.

[4] Andrew Gelman, John B Carlin, Hal S Stern, and Donald B Rubin. *Bayesian data analysis*. Chapman and Hall/CRC, 1995.

[5] Scott M Berry, C Shane Reese, and Patrick D Larkey. Bridging different eras in sports. *Journal of the American Statistical Association*, 94(447):661–676, 1999.

[6] Geoffroy Berthelot, Stéphane Len, Philippe Hellard, Muriel Tafflet, Marion Guillaume, Jean-Claude Vollmer, Bruno Gager, Laurent Quinquis, Andy Marc, and Jean-François Toussaint. Exponential growth combined with exponential decline explains lifetime performance evolution in individual and human species. *Age*, 34(4):1001–1009, 2012.

[7] James A Brander, Edward J Egan, and Louisa Yeung. Estimating the effects of age on nhl player performance. *Journal of Quantitative Analysis in Sports*, 10(2):241–259, 2014.

[8] John Charles Bradbury. Peak athletic performance and ageing: evidence from baseball. *Journal of Sports Sciences*, 27(6):599–610, 2009.

[9] Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. Fitting linear mixed-effects models using lme4. *Journal of statistical software*, 67:1–48, 2015.

[10] SJ Dormehl, SJ Robertson, and CA Williams. Modelling the progression of male swimmers' performances through adolescence. sports. 2016; 4 (1): 2.

[11] Jim E Griffin, Laurenţiu C Hinoveanu, and James G Hopker. Bayesian modelling of elite sporting performance with large databases. *Journal of Quantitative Analysis in Sports*, 18(4):253–268, 2022.

[12] M Spyropoulou, JG Hopker, and JE Griffin. Modelling between-and within-season trajectories in elite athletic performance data. *arXiv preprint arXiv:2405.17214*, 2024.

[13] William Hopkins, Stephen Marshall, Alan Batterham, Juri Hanin, et al. Progressive statistics for studies in sports medicine and exercise science. *Medicine+ Science in Sports+ Exercise*, 41(1):3, 2009.

[14] Mike West and Jeff Harrison. *Bayesian forecasting and dynamic models*. Springer, 1997.

[15] Jeff T Terpstra and Nicholas D Schauer. A simple random walk model for predicting track and field world records. *Journal of Quantitative Analysis in Sports*, 3(3), 2007.

[16] Roderick JA Little and Donald B Rubin. *Statistical analysis with missing data*. John Wiley & Sons, 2019.

[17] Jill Borresen and Michael Ian Lambert. The quantification of training load, the training response and the effect on performance. *Sports medicine*, 39(9):779–795, 2009.

[18] Stan Development Team. Stan user's guide: Multivariate hierarchical priors. `https://mc-stan.org/docs/stan-users-guide/regression.html#multivariate-hierarchical-priors.section`.

[19] Daniel Lewandowski, Dorota Kurowicka, and Harry Joe. Generating random correlation matrices based on vines and extended onion method. *Journal of multivariate analysis*, 100(9):1989–2001, 2009.

[20] Stephen C Hollings, Will G Hopkins, and Patria A Hume. Age at peak performance of successful track & field athletes. *International Journal of Sports Science & Coaching*, 9(4):651–661, 2014.

[21] Andrew Gelman and Jennifer Hill. *Data analysis using regression and multilevel/hierarchical models*. Cambridge university press, 2007.

[22] Joram Soch et al. The Book of Statistical Proofs (StatProofBook). Zenodo, 2024.

[23] Havard Rue and Leonhard Held. *Gaussian Markov random fields: theory and applications*. Chapman and Hall/CRC, 2005.

[24] Aki Vehtari, Andrew Gelman, and Jonah Gabry. Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and computing*, 27(5):1413–1432, 2017.

[25] Tianfeng Chai, Roland R Draxler, et al. Root mean square error (rmse) or mean absolute error (mae). *Geoscientific model development discussions*, 7(1):1525–1534, 2014.

[26] Alan E Gelfand and Sujit K Ghosh. Model choice: a minimum posterior predictive loss approach. *Biometrika*, 85(1):1–11, 1998.

[27] Aki Vehtari and Janne Ojanen. A survey of bayesian predictive methods for model assessment, selection and comparison. 2012.

[28] Aditi S Majumdar and Robert A Robergs. The science of speed: Determinants of performance in the 100 m sprint. *International Journal of Sports Science & Coaching*, 6(3):479–493, 2011.

[29] Roel Vaeyens, Matthieu Lenoir, A Mark Williams, and Renaat M Philippaerts. Talent identification and development programmes in sport: current models and future directions. *Sports medicine*, 38(9):703–714, 2008.

[30] Jason Gulbin, Juanita Weissensteiner, Karen Oldenziel, and Françoys Gagné. Patterns of performance development in elite athletes. *European journal of sport science*, 13(6):605–614, 2013.

[31] World Athletics. Men's 100 metres season top list (outdoor, senior, 2025) — page 3. `https://worldathletics.org/records/toplists/sprints/100-metres/outdoor/men/senior/2025?page=3`, 2025.

[32] World Athletics. Men's 100 metres final – results, world athletics championships oregon 2022. `https://worldathletics.org/results/world-athletics-championships/2022/world-athletics-championships-oregon-2022-7137279/men/100-metres/final/result`, 2022.

[33] Olympic Games. Olympic champion marcell jacobs withdraws from track & field worlds 100m semi-finals. https://www.olympics.com/en/news/olympic-champion-marcell-jacobs-withdraws-from-track-field-worlds-100m-semi-fina, 2022. Accessed: 2025-08-24.

[34] World Athletics. World athletics statistics handbook. http://www.worldathletics.org/stats-zone, 2022. Accessed: 2025-08-24.

[35] Adelchi Azzalini and Antonella Capitanio. Distributions generated by perturbation of symmetry with emphasis on a multivariate skew t-distribution. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 65(2):367–389, 2003.

[36] Bob Carpenter, Andrew Gelman, Matthew D Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of statistical software*, 76:1–32, 2017.

Appendix

A.1 Random Effect STAN model

```
data {
  int<lower=1> N;          // number of athletes
  int<lower=1> J;          // number of unique ages
  real y[N, J];            // Observed outcomes: each row is an athlete
  vector[J] unique_ages;   // The sorted vector of unique ages
  int<lower=0, upper=1> missing_positions[N, J];
}
parameters {
  // LKJ prior for the full J x J correlation matrix.
  corr_matrix[J] Omega;
  // Standard deviations for the random effects, prior scale
  vector<lower=0>[J] sigma;
  // Subject-specific random effects
  matrix[N, J] gamma;
  // Residual error standard deviation
  real<lower=0> sigma_y;
  // Population mean effects
  vector[J] gamma_0;
}
transformed parameters {
  // Covariance matrix for random effects
  cov_matrix[J] Sigma;
  Sigma = quad_form_diag(Omega, sigma);
}
model {
  // Priors on covariance parameters and residual noise
  sigma ~ cauchy(0, 2.5);
  Omega ~ lkj_corr(1);
  sigma_y ~ cauchy(0, 2.5);
  gamma_0 ~ normal(0, 10^3);

   // Prior for the random effects
  for (n in 1:N) {
    gamma[n] ~ multi_normal(gamma_0, Sigma);
  }

  // Likelihood
  for (n in 1:N) {
    for (j in 1:J) {
      if (!missing_positions[n, j]) {  // Only model if not missing
        y[n, j] ~ normal(gamma[n, j], sigma_y);
      }
    }
  }
}
```

A.2 Fixed Effect STAN model

```
data {
  int<lower=1> N;                  // number of athletes
  int<lower=1> J;                  // number of unique ages
  int<lower=1> max_obs;            // maximum number of observed components
     across all athletes
  int<lower=1> obs_count[N];
  int<lower=1, upper=J> age_idx[N, max_obs];
  vector[J] unique_ages;           // The sorted vector of unique ages
  real centre_age;                 // age to centre the polynomial
  real y[N, max_obs];              // matrix of observations for each subject
}

transformed data {
  vector[J] centred_ages = unique_ages - centre_age;
  matrix[J, 5] X_poly;
  for (j in 1:J) {
    X_poly[j, 1] = 1;
    X_poly[j, 2] = centred_ages[j];
    X_poly[j, 3] = square(centred_ages[j]);
    X_poly[j, 4] = centred_ages[j]^3;
    X_poly[j, 5] = centred_ages[j]^4;
  }
}

parameters {
  vector[5] beta;
  vector<lower=0>[J] sigma;
  corr_matrix[J] Omega;
}

transformed parameters {
  // Covariance matrix for random effects
  cov_matrix[J] Sigma;
  Sigma = quad_form_diag(Omega, sigma);
}

model {
  sigma ~ cauchy(0, 2.5);
  Omega ~ lkj_corr(1);
  beta ~ normal(0, 100);

  for (n in 1:N) {
    int k = obs_count[n];
    vector[k] mu;
    matrix[k, k] Sigma_n;

    for (i in 1:k) {
      mu[i] = dot_product(row(X_poly, age_idx[n, i]), beta);
      for (j in 1:k) {
        Sigma_n[i, j] = Sigma[age_idx[n, i], age_idx[n, j]];
      }
    }
    vector[k] y_obs_n;
    for (i in 1:k) {
      y_obs_n[i] = y[n, i];
    }
    y_obs_n ~ multi_normal(mu, Sigma_n);
  }
}
```

A.3 Combined Effect STAN model

```
data {
  int<lower=1> N;
  int<lower=1> J;
  int<lower=1> max_obs;
  int<lower=1> obs_count[N];
  int<lower=1, upper=J> age_idx[N, max_obs];
  vector[J] unique_ages;
  real centre_age;
  real y[N, max_obs];
}

transformed data {
  vector[J] centred_ages = unique_ages - centre_age;
  matrix[J, 5] X_poly;
  for (j in 1:J) {
    X_poly[j, 1] = 1;
    X_poly[j, 2] = centred_ages[j];
    X_poly[j, 3] = square(centred_ages[j]);
    X_poly[j, 4] = centred_ages[j]^3;
    X_poly[j, 5] = centred_ages[j]^4;
  }
}

parameters {
  vector[5] beta;
  vector[J] gamma[N];
  vector<lower=0>[J] sigma;
  corr_matrix[J] Omega;
  real<lower=0> sigma_y;
}

transformed parameters {
  cov_matrix[J] Sigma;
  Sigma = quad_form_diag(Omega, sigma);
}

model {
  sigma ~ cauchy(0, 2.5);
  Omega ~ lkj_corr(1);
  beta ~ normal(0, 100);
  sigma_y ~ cauchy(0, 2.5);

  for (n in 1:N) {
    gamma[n] ~ multi_normal(rep_vector(0, J), Sigma);
  }

  for (n in 1:N) {
    int k = obs_count[n];
    vector[k] mu;
    vector[k] y_obs_n;

    for (i in 1:k) {
      int a_idx = age_idx[n, i];
      mu[i] = dot_product(row(X_poly, a_idx), beta) + gamma[n][a_idx];
      y_obs_n[i] = y[n, i];
    }
    y_obs_n ~ normal(mu, sigma_y);
  }
}
```

A.4 1st Order Random Walk STAN model

```
data {
  int<lower=1> N;
  int<lower=1> J;
  int<lower=1> max_obs;
  int<lower=1> obs_count[N];
  int<lower=1, upper=J> age_idx[N, max_obs];
  vector[J] unique_ages;
  real centre_age;
  real y[N, max_obs];
}

transformed data {
  vector[J] centred_ages = unique_ages - centre_age;
  matrix[J, 5] X_poly;
  for (j in 1:J) {
    X_poly[j, 1] = 1;
    X_poly[j, 2] = centred_ages[j];
    X_poly[j, 3] = square(centred_ages[j]);
    X_poly[j, 4] = centred_ages[j]^3;
    X_poly[j, 5] = centred_ages[j]^4;
  }
}

parameters {
  vector[5] beta;                  // fixed effects for age polynomial
  matrix[N, J] gamma;              // athlete-specific deviations
  real<lower=0> sigma;             // SD for initial gamma
  real<lower=0> tau;               // SD for random walk increments
  real<lower=0> sigma_y;           // residual SD
}

model {
  // Priors
  beta ~ normal(0, 100);
  sigma ~ cauchy(0, 2.5);
  tau ~ cauchy(0, 2.5);
  sigma_y ~ cauchy(0, 2.5);

  // Random walk prior for gamma
  for (n in 1:N) {
    gamma[n, 1] ~ normal(0, sigma);  // Initial deviation
    for (j in 2:J) {
      gamma[n, j] ~ normal(gamma[n, j-1], tau);  // Random walki increments
    }
  }

  for (n in 1:N) {
    int k = obs_count[n];
    vector[k] mu;
    vector[k] y_obs_n;

    for (i in 1:k) {
      int a_idx = age_idx[n, i];
      mu[i] = dot_product(row(X_poly, a_idx), beta) + gamma[n, a_idx];
      y_obs_n[i] = y[n, i];
    }
    y_obs_n ~ normal(mu, sigma_y);
  }
}
```

A.5 2nd Order Random Walk STAN model

```
data {
  int<lower=1> N;
  int<lower=1> J;
  int<lower=1> max_obs;
  int<lower=1> obs_count[N];
  int<lower=1, upper=J> age_idx[N, max_obs];
  vector[J] unique_ages;
  real centre_age;
  real y[N, max_obs];
}

transformed data {
  vector[J] centred_ages = unique_ages - centre_age;
  matrix[J, 5] X_poly;
  for (j in 1:J) {
    X_poly[j, 1] = 1;
    X_poly[j, 2] = centred_ages[j];
    X_poly[j, 3] = square(centred_ages[j]);
    X_poly[j, 4] = centred_ages[j]^3;
    X_poly[j, 5] = centred_ages[j]^4;
  }
}

parameters {
  vector[5] beta;
  matrix[N, J] gamma;
  real<lower=0> sigma_1;
  real<lower=0> sigma_2;
  real<lower=0> tau;
  real<lower=0> sigma_y;
}

model {
  beta ~ normal(0, 100);
  sigma_1 ~ cauchy(0, 2.5);
  sigma_2 ~ cauchy(0, 2.5);
  tau ~ cauchy(0, 2.5);
  sigma_y ~ cauchy(0, 2.5);

  // Random walk prior for gamma
  for (n in 1:N) {
    gamma[n, 1] ~ normal(0, sigma_1);
    gamma[n, 2] ~ normal(0, sigma_2);
    for (j in 3:J) {
      gamma[n, j] ~ normal(2*gamma[n, j-1]-gamma[n, j-2], tau);  // RW
          increments
    }
  }

  // Likelihood
  for (n in 1:N) {
    int k = obs_count[n];
    vector[k] mu;
    vector[k] y_obs_n;

    for (i in 1:k) {
      int a_idx = age_idx[n, i];
      mu[i] = dot_product(row(X_poly, a_idx), beta) + gamma[n, a_idx];
      y_obs_n[i] = y[n, i];
```

```
    }
    y_obs_n ~ normal(mu, sigma_y);
  }
}
```

A.6 Posterior covariance matrix for Male 100 m

Age	17	18	19	20	21	22	23	24	25	26	27	28	29	30
17	0.0321	0.0249	0.0155	0.0143	0.0117	0.00842	0.0118	0.01199	0.01153	0.01013	0.00676	0.00912	-0.000667	0.0124
18	0.0249	0.0737	0.0225	0.0205	0.0163	0.0154	0.0139	0.0146	0.00995	0.00992	0.00577	0.01299	0.00593	0.0128
19	0.0155	0.0225	0.0367	0.0220	0.0211	0.0179	0.0159	0.0150	0.0105	0.0125	0.00687	0.0155	0.00792	0.0141
20	0.0143	0.0205	0.0220	0.0283	0.0229	0.0222	0.0182	0.0192	0.0164	0.0154	0.0153	0.0145	0.0103	0.0151
21	0.0117	0.0163	0.0211	0.0229	0.0398	0.0249	0.0249	0.0210	0.0178	0.0176	0.0155	0.0161	0.0154	0.0160
22	0.00842	0.0154	0.0179	0.0222	0.0249	0.0378	0.0265	0.0229	0.0207	0.0181	0.0180	0.0181	0.0168	0.0165
23	0.0118	0.0139	0.0159	0.0182	0.0249	0.0265	0.0557	0.0294	0.0245	0.0214	0.0220	0.0184	0.0220	0.0236
24	0.01199	0.0146	0.0150	0.0192	0.0210	0.0229	0.0294	0.0465	0.0371	0.0252	0.0222	0.0217	0.0217	0.0229
25	0.01153	0.00995	0.0105	0.0164	0.0178	0.0207	0.0245	0.0371	0.0602	0.0371	0.0328	0.0379	0.0291	0.0386
26	0.01013	0.00992	0.0125	0.0154	0.0176	0.0181	0.0214	0.0252	0.0371	0.0556	0.0321	0.0281	0.0284	0.0219
27	0.00676	0.00577	0.00687	0.0153	0.0155	0.0180	0.0220	0.0222	0.0328	0.0321	0.0662	0.0311	0.0293	0.0278
28	0.00912	0.01299	0.0155	0.0145	0.0161	0.0181	0.0184	0.0217	0.0379	0.0281	0.0311	0.0852	0.0395	0.0290
29	-0.000667	0.00593	0.00792	0.0103	0.0154	0.0168	0.0220	0.0217	0.0291	0.0284	0.0293	0.0395	0.0858	0.0410
30	0.0124	0.0128	0.0141	0.0151	0.0160	0.0165	0.0236	0.0229	0.0386	0.0219	0.0278	0.0290	0.0410	0.0462

Table 7: Posterior mean covariance matrix $\mathbf{\Sigma}$ (ages 17-30)

A.7 Additional results for Female 100 m

Competitor	**Nat**	**Prob. Final**	**Stage Reached**
Elaine Thompson-Herah	JAM	0.539	Final
Twanisha Terry	USA	0.420	Semi-Final
Dina Asher-Smith	GBR	0.295	Final
Tamara Clark	USA	0.285	DNC
Teahna Daniels	USA	0.249	DNC
Ty'Nia Gaither	BAH	0.217	Semi-Final
Dezerea Bryant	USA	0.206	DNC
Gina Bass	GAM	0.201	DNC

Table 8: Top 8 predicted probabilities of reaching 2022 final with actual competition outcome for female athletes